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Question Paper Code : 21568

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2024

Third Semester

Computer Science and Engineering

23CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2023)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A, CO1 1. In 16-bit 2's complement representation, the decimal number -28 is_____. Justify your answer.

A) 1000 0000 1110 0100 B) 0000 0000 1110 0100

C) 1111 1111 1110 0100 D) 1111 1111 0001 1100

R, CO1 2. State De Morgan's theorem.

A, CO2 3. A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The minimum number of select lines needed for the multiplexer is _____. Justify your answer.

A, CO2 4. Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed? Justify your answer.

A) 2^n line to 1 line B) 2^{n+1} line to 1 line

C) $^{2n-1}$ line to 1 line D) 2^{n-2} line to 1 line

R, CO3 5. In a DRAM _____ . Comment on your answer.

A) Periodic refreshing is not required

B) Information is stored in a capacitor

C) Information is stored in a latch

D) Both read and write operations can be performed simultaneously

R, CO3 6. Distinguish between decoder and encoder.

U, CO4 7. We want to design a synchronous counter that counts the sequence 0−1−0−2−0−3 and then repeat. The minimum number of J−K flip-flops required to implement this counter is _____. Justify your answer.

U, CO4 8. Give the comparison between combinational circuits and sequential circuits.

R, CO5 9. What is a critical and non-critical race?

U, CO5 10. What is meant by hazard and how it could be avoided?

PART B – (4 × 15 = 60 Marks)

A, CO1 11. a) Construct a Karnaugh Map for the following function $F(A,B,C,D,E)=\Sigma_m(0,5,6,8,9,10,11,16,20,24,25,26,27,29,31)$ and draw the logic diagram using NAND gates. (15 Marks)

OR

A, CO1 11. b) Reduce the following equation using Quine McClucky method of minimization $F(A,B,C,D) = \Sigma m(0,1,3,4,5,7,10,13,14,15)$ and Implement the same. (15 Marks)

A, CO2 12. a) Construct a combinational circuit to convert a given binary coded decimal number into an Excess 3 code for example when the input to the gate is 0110 then the circuit should generate output as1001. (15 Marks)

OR

A, CO2 12. b) i. Design a 2 bit magnitude comparator and explain its operation in detail. (8 Marks)

A, CO2 12. b) ii. Implement the following Boolean function using 4 x 1 multiplexers. $F=\Sigma_m(1,2,3,6,7,8,11,12,14)$ (7 Marks)

A, CO3 13. a) i. Design a PLA structure for the following functions.

$F1 = \Sigma m(5,7,10,13,14,15)$.

$F2 = \Sigma m(0,2,5,7,8,10,14,15)$. (8 Marks)

U, CO3 13. a) ii. Discuss the classification of RAM and ROM memories illustrating with its Block diagrams. (7 Marks)

OR

A, CO3 13. b) Using 4×3×4 PALs, determine the circuit used to realize the following Boolean functions.

$W(A,B,C,D) = \sum(0,2, 6,7,8,9,12,13)$

$X(A,B,C,D) = \sum(0, 2, 6, 7, 8, 9, 12, 13, 14)$

$Y(A, B, C, D) = \sum(2, 3, 8, 9, 10, 12, 13)$ (15 Marks)

A, CO4	14. a) i.	Realize SR flip-flop using D flip-flop and JK flip-flop.	(8 Marks)
A, CO4	14. a) ii.	Write a HDL code to design a SISO shift register.	(7 Marks)

OR

A, CO4	14. b)	With a neat state diagram, design and explain the working of a synchronous mod – 12 counter.	(15 Marks)

PART C – (1 × 20 = 20 Marks)

A, CO5	15. a)	Design a circuit with inputs X and Y to give an output Z= 1 when XY=11 but only if X becomes 1 before Y, by drawing total state diagram, primitive flow table and output map in which transient state is included.	(20 Marks)

OR

A, CO5	15. b)	Explain Hazards and its types in detail. Check whether the circuit given by $Y= x_1 x_2 + x_2'x_3$ contain hazards. If the hazard is present, demonstrate its removal.	(20 Marks)